The language of genetics

Many scientific terms have their origins in Greek or Latin words. The list below shows some terms used in genetics, their origins (Greek or Latin) and their meanings.

1 Draw an asterisk (*) in front of the terms in bold that are nouns, and underline the verbs.

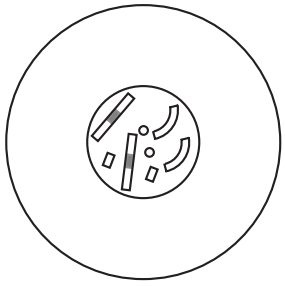
- **allellon** (Greek) *mutual, other*
- **allos** (Greek) *other*
- **chroma** (Greek) *colour*
- **dominare** (Latin) *to rule, to be master*
- **filia** (Latin) *daughter*
- **filius** (Latin) *son*
- **genea** (Greek) *breed*
- **generare** (Latin) *to produce*
- **genesis** (Greek) *origin, creation*
- **genesis** (Latin) *birth, creation*
- **genos** (Greek) *origin, race*
- **helix** (Greek) *spiral, twisted thing*
- **heteros** (Greek) *other*
- **homos** (Greek) *same*
- **hybrida** (Latin) *mongrel*
- **kytos** (Greek) *hollow vessel*
- **morphe** (Greek) *shape*
- **nucleus** (Latin) *kernel, nut*
- **parens** (Latin) *mother or father*
- **phainein** (Greek) *to show*
- **plasma** (Greek) *something moulded or formed*
- **recessus** (Latin) *retreat, a going back*
- **soma** (Greek) *body*
- **typos** (Greek) *model, image, form*
- **typus** (Latin) *image, form*
- **zygotes** (Greek) *yoked, joined*

2 Use the information above to help you write a definition for each of these scientific terms.

a allele (short for allelomorph)
b chromosome
c cytoplasm
d dominant
e gene
f F generation or filia generation
g genotype
h helix
i heterozygous alleles
j homozygous alleles
k hybrid
l nucleotide
m nucleus
n P generation or parent generation
o phenotype
p recessive
q zygote

3 Use terms from question 2 to label the diagram of a cell.

4 Look at the chromosomes in the cell diagram and label the nucleus as heterozygous or homozygous.



5 Is a heterozygous cell about genotype or phenotype?

6 Identify two alleles from these genes: green eye colour, tall body size, brown hair colour, short body size.

7 An animal has a gene for black fur and a gene for white fur, and looks black. State:

a the phenotype.
b the genotype.
c the dominant gene.
d the recessive gene.
e the alleles.
f whether the animal is a hybrid.
g whether the animal is heterozygous or homozygous.

8 Use terms from question 2 to complete the text.

The control centre of a cell is the _____, which is surrounded by _____. Inside it there are long, thread-like, dark-coloured structures called _____. Each chromosome is made up of the chemical DNA, which looks like a double _____. Each DNA molecule is made up of smaller units called _____. There are only four different nucleotides; the DNA molecules are different because they have different amounts of nucleotides and the nucleotides are in different sequences. Several _____ together represent a gene and all the _____ together make up a _____. The lengths and numbers of chromosomes vary for different organisms.

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